

I need to quantify the overconfidence of company Boards in assessing the effectiveness and accuracy of their decisions and actions. This will be for a Bayesian model, so I am interested in the observed range of overconfidence effects, and I want to quantify bias in mean (overestimating effectiveness) and bias in variance (overestimating certainty of outcome).

My specific context is shareholder votes against executive pay. A Board is considering commissioning an external review of governance as a way to reduce the upcoming protest votes. But you can use any research that is relevant.

Your task is to

1. do a synthesis of the relevant research, provide a descriptive summary
2. then recommend ranges for Board overconfidence effect in the mean and variance.

Synthesis: Board Overconfidence and Governance Decisions

Research on board and managerial overconfidence shows that boards systematically **overestimate the quality and consequences** of their strategic and governance decisions, and are poorly calibrated about this.

Evidence of Board-level Overconfidence

Board overconfidence has been most directly studied in mergers and acquisitions (M&A). U.S. and U.K. evidence shows that boards that have recently been “successful” acquirers become overconfident via biased self-attribution, pay higher premiums, and generate **poorer subsequent deal performance and weaker investor reactions** than less overconfident boards (Twardawski & Kind, 2023; Brahma et al., 2023). Overconfidence is proxied by repeated acquisitions, cash financing, and board composition; these variables explain economically meaningful differences in returns (Twardawski & Kind, 2023; Brahma et al., 2023).

In Chinese listed firms, overconfidence **contagion** from the chair to directors is documented: directors “blindly” follow an overconfident chair through a voting mechanism, and this collective overconfidence leads to distorted investment decisions (Liang et al., 2021). The effect is stronger in more authoritarian, state-owned enterprises and is mitigated by more board meetings and discussion (Liang et al., 2021).

An experimental study using real board members in a hidden-profile decision task finds that only about **20% of boards choose the objectively best option**, yet members report high satisfaction and perceive that they weighed pros and cons adequately (Coffeng et al., 2021). This dissociation between poor objective quality and positive subjective evaluations indicates boards are **unaware of their biased decision-making** and overconfident in process and outcome (Coffeng et al., 2021).

Broader governance work shows that managerial overconfidence weakens the effectiveness of governance mechanisms and is associated with lower firm performance; overconfidence negatively moderates the performance benefits of board independence and ownership concentration (Guluma, 2021). Systematic reviews confirm that overconfidence is a pervasive managerial bias affecting investment, financing, and reporting choices (Chanyal, 2025; Karki et al., 2024).

Shareholder Voting, Executive Pay, and Board Miscalibration

Shareholder dissent on pay and other governance items functions as an **external correction to board misjudgment**. Systematic reviews of say-on-pay and pay-related activism show that dissent is triggered not only by raw pay levels but by perceptions of “excess” pay and fairness, inconsistent with purely rational principal-agent models (Obermann & Velte, 2018). In a large sample of “vote-no” campaigns and pay-related proposals, firms with **excess CEO pay** that are targeted subsequently cut CEO pay by about **\$7.3m on average**, indicating boards had previously set pay at levels shareholders viewed as materially misaligned (Ertimur et al., 2011).

More generally, shareholder proposals and protest votes are concentrated in firms with poor performance and weak governance, and successful proposals lead to measurable governance improvements (e.g., reduced earnings management, stronger monitoring) (Renneboog & Szilagyi, 2011; Fan & Radhakrishnan, 2020; Li et al., 2019). A regression-discontinuity design around close shareholder-proposal votes shows that marginally passing proposals produce **discrete reductions in earnings management**, implying that pre-intervention boards were overconfident in the acceptability of their reporting practices and governance arrangements (Fan & Radhakrishnan, 2020; Li et al., 2019).

In director-pay contexts, interview evidence suggests boards and remuneration committees tend to view their pay policies as fair and well justified, while only **substantial protest votes** or regulatory “two-strikes” mechanisms force visible change, again consistent with prior overconfidence about shareholder acceptance (Van Zyl & Mans-Kemp, 2022).

Metacognitive Insights for Mean vs. Variance Overconfidence

Cognitive models of confidence provide quantitative structure useful for a Bayesian specification. Confidence can be modeled as a noisy estimate of decision reliability, limited by “meta-uncertainty” (uncertainty about one’s own uncertainty) (Boundy-Singer et al., 2022). Parametric models of metacognitive readout show that:

- Confidence reports are **systematically biased**, not just noisy, often shifted upward relative to true accuracy (Boundy-Singer et al., 2022; Guggenmos, 2021).
- There is additional **second-order noise** in the mapping from internal evidence to confidence (“metacognitive noise”), which can be estimated separately from first-order decision noise (Guggenmos, 2021).

These frameworks naturally map to a Bayesian model in which boards specify an internal prior and likelihood (their governance view), but their reported confidence is an **over-precise and biased transformation** of that internal distribution.

Recommended Parameter Ranges for a Bayesian Model

The literature does not directly estimate numeric “overconfidence parameters” for boards around pay votes, so any calibration must be stylized and anchored to observed effect sizes and miscalibration patterns.

The following ranges are consistent with:

- Large economic mistakes in M&A attributable to overconfident boards (Twardawski & Kind, 2023; Brahma et al., 2023).
- Low objective accuracy vs. high subjective satisfaction in board decision experiments (Coffeng et al., 2021).
- Material ex-post governance corrections only after strong shareholder pressure (Ertimur et al., 2011; Fan & Radhakrishnan, 2020; Li et al., 2019; Van Zyl & Mans-Kemp, 2022).
- Human metacognitive miscalibration ranges from psychology (Boundy-Singer et al., 2022; Guggenmos, 2021).

Interpret these as priors for a “typical” overconfident board in a setting like anticipated say-on-pay dissent.

Suggested Mean-Bias Calibration (Overestimation of Effectiveness)

Define μ^* as the true expected reduction in dissent from an action (e.g., commissioning an external review), and $\hat{\mu}$ as the board's ex-ante believed effect (in percentage-point reduction in "against" or "abstain" votes).

A reasonable parametric bias:

- **Multiplicative mean overconfidence:**
 $\hat{\mu} = (1 + \beta) \mu^*$, with $\beta \sim \text{Uniform}(0.25, 1.0)$

This implies boards on average **overestimate effectiveness by 25–100%**. Examples:

- If the true effect is a 5-point reduction in dissent, boards expect 6.25–10 points.
- For discrete binary outcomes ("will this avert a visible protest?"), this corresponds roughly to board-perceived success probabilities **10–25 percentage points higher** than is justified (e.g., believing 70–80% success when the true probability is 50–60%), consistent with typical human overconfidence in moderate-difficulty domains (Boundy-Singer et al., 2022; Guggenmos, 2021) and the magnitude of ex-post corrections in pay and M&A settings (Twardawski & Kind, 2023; Brahma et al., 2023; Ertimur et al., 2011; Fan & Radhakrishnan, 2020; Coffeng et al., 2021).

If a multiplicative form is inconvenient, an additive bias such as $\hat{\mu} = \mu^* + \delta$ with $\delta \sim \text{Normal}(0, 0.05^2)$ in vote-share units (i.e., typical upward error around **5 percentage points**) is also plausible, given that meaningful say-on-pay protest thresholds are often in the 10–30% dissent range (Ertimur et al., 2011; Obermann & Velte, 2018; Yermack, 2010).

Suggested Variance-Bias Calibration (Over-certainty about Outcomes)

Let σ^{*2} be the true variance in the effect of the board's action (between-world uncertainty about reduction in dissent), and $\hat{\sigma}^2$ the variance implied by the board's expressed confidence.

Empirical metacognition work and the hidden-profile board experiment jointly suggest that decision-makers are **too sure** of their judgments; errors are common but not anticipated (Boundy-Singer et al., 2022; Guggenmos, 2021; Coffeng et al., 2021). A stylized variance bias:

- **Underestimation of variance (over-precision):**
 $\hat{\sigma}^2 = \sigma^{*2} / \kappa$, with $\kappa \sim \text{Uniform}(2, 5)$

So boards act as if outcomes are **2–5x more precise** than they really are. In probability-space:

- If the true standard deviation of the vote-share effect is 4 points, the board behaves as if it were 1.8–2.8 points.
- In binary success/failure terms, this corresponds to **95% credible intervals** on success probability that are about $\sqrt{\kappa} \approx 1.4\text{--}2.2$ **times too narrow**.

To connect more directly with metacognitive models, you could instead treat the board's reported confidence C as being derived from an internal log-odds L through a biased link:

- $\hat{L} = \alpha + \gamma L + \varepsilon_{\text{meta}}$, with $\gamma > 1$ indicating over-precision and $\varepsilon_{\text{meta}} \sim \text{Normal}(0, \tau^2)$ a meta-uncertainty term (Boundy-Singer et al., 2022; Guggenmos, 2021).

Plausible parameter ranges inspired by human data:

- $\alpha \in [0, 0.5]$ (small positive shift toward higher confidence)
- $\gamma \in [1.2, 1.8]$ (20–80% steeper mapping than calibrated)
- τ in the same order as the decision noise SD, capturing substantial metacognitive noise (Boundy-Singer et al., 2022; Guggenmos, 2021).

Practical Implementation for Your Context

For a board considering an external governance review to reduce protest votes on pay, a simple, transparent Bayesian specification could be:

- True effect $\mu^* \sim \text{Normal}(m_0, s_0^2)$, e.g., $m_0 = -0.05$ (5-point reduction in dissent), $s_0 = 0.04$.
- Board's internal belief before overconfidence distortion: same μ^* .
- Reported belief used for their decision:
 - $\hat{\mu} = \mu^* (1 + \beta)$, $\beta \sim \text{Uniform}(0.25, 1.0)$
 - $\hat{\sigma}^2 = \sigma^{*2} / \kappa$, $\kappa \sim \text{Uniform}(2, 5)$

You can then explore:

- How often an overconfident board decides **not** to commission a review because it overestimates the effect of less costly or symbolic changes on protest votes.
- How sensitive outcomes are to reducing β and increasing κ (e.g., via prior shareholder dissent, external advisors, or board diversity mechanisms that weaken overconfidence contagion (Liang et al., 2021; Gurdgiev & Ni, 2023; Guluma, 2021; Van Zyl & Mans-Kemp, 2022)).

These ranges are deliberately conservative: they are large enough to match observed governance failures and corrections, but not extreme relative to human confidence miscalibration observed in cognitive experiments and in real boards.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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